

Given the code fragment:

```
public static void main(String[] args) {  
    String name = "Rita";  
    int age = 14;  
    /* line n1 */  
}
```

Which code fragment, when inserted at line n1, enables it to print Rita is 14 years old?

- ☐ A. `System.out.println("%s is %d years old" +name+age);`
- ☐ B. `System.out.println("%s is %n years old" name, age);`
- ☐ C. `System.out.printf("%s is %d years old", name, age);`
- ☐ D. `System.out.printf("%s is %n years old", name, age);`

The answer is: C

Given the code fragment:

```
String flavors[] = {"Vanilla", "Chocolate"};  
int choice = 2;  
switch(choice) {  
    case 1:  
        System.out.println("Selected " + flavors[1] + " flavor.");  
        break;  
    case 2:  
        System.out.println("Selected " + flavors[2] + " flavor.");  
        break;  
    default:  
        System.out.println("Thank you!");  
}
```

What is the result?

- ☐ A. Selected null flavor.
- ☐ B. Selected Chocolate flavor.
- ☐ C. Selected Chocolate flavor.
- ☐ D. An `ArrayIndexOutOfBoundsException` is thrown at run time.

The answer is: D

Given the code fragment:

```
String digits = "0123456789";  
System.out.println(  
    digits.substring(  
        digits.indexOf ("4"), digits.indexOf ("8")).concat ("89"));
```

What is the result?

- ☐ A. 3456789
- ☐ B. 45678
- ☐ C. 4567889
- ☐ D. 456789

The answer is: D

Which method identifier is correct according to Java naming conventions?

- ☐ A. calculateBill
- ☐ B. Calculator
- ☐ C. calculatebill
- ☐ D. BillCalculator

The Answer is: C

Given the code fragment:

```
class Ball {  
    double weight;  
}  
public class App {  
    public static void main(String[] args) {  
        //line n1  
        System.out.println(b.weight);  
    }  
}
```

Which code fragment can be inserted at line n1 to enable the code to print 0.0?

- ☐ A. Ball.weight = 0.0;
- ☐ B. Ball b = new Ball(0.0);
- ☐ C. Ball b = null;
b.weight = 0.0;
- ☐ D. Ball b = new Ball();

The answer is: D

Given the contents of the Test.java file:

```
class MyClass {  
    private int var1 = 100;  
    public int var2 = 200;  
    public void doCalc() {  
        var1 = 100 * 2; // line n1  
        var2 = 200 * 2;  
    }  
}  
  
public class Test {  
    public static void main(String[] args) {  
        MyClass t = new MyClass();  
        t.doCalc();  
        System.out.println (t.var1 + " : " + t.var2); // line n2  
    }  
}
```

What is the result?

- ☐ A. A compilation error occurs at line n1.
- ☐ B. 200 : 400
- ☐ C. 100 : 400
- ☐ D. A compilation error occurs at line n2.

The answer is: D

Given:

```
class Bus {  
    String type = "default";  
    // line n1  
    Bus (String type) {  
        // line n2  
        this.type = type;  
    }  
}  
  
public class App{  
    public static void main(String[] args) {  
        Bus b1 = new Bus();  
        System.out.println(b1.type);  
        Bus b2 = new Bus("luxury");  
        System.out.println (b2.type);  
    }  
}
```

What is the result?

- ☐ A. The code fails to compile. To make it compile, at line n1 insert:
this() {}
- ☐ B. The code fails to compile. To make it compile, at line n2 insert:
this();
- ☐ C. The code fails to compile. To make it compile, at line n1 insert:
Bus() {}
- ☐ D. The code compiles and prints:
default

The answer is: C

You have a microprocessor board, such as Raspberry PI, wired to control a drone. Which edition of Java is geared towards use of simple, closed systems with constrained memory requirements, such as a microprocessor board?

- ☐ A. Java Enterprise Edition
- ☐ B. Java Standard Edition with a Compact Profile
- ☐ C. Java Micro Edition
- ☐ D. Java SE Embedded

The answer is: C

Which statement is true about primitive variables?

- ☐ A. They can be compared with the equals method only.
- ☐ B. They can be compared with the compareTo method only.
- ☐ C. They cannot be compared.
- ☐ D. They can be compared with the == operator.

The answer is: D

Given the code fragment:

```
String a = "Java";  
String b = new String ("Java");  
System.out.println (a.equals(b));  
System.out.println (a==b);
```

What is the result?

☐ **A. true**

true

☐ **B. true**

false

☐ **C. false**

false

☐ **D. false**

true

The answer is: B